

PROBABILITY AND STATISTICS
RANDOM VARIABLES, DISTRIBUTIONS
TUTORIAL SHEET

1. The CDF of a random variable X is given by $F_X(x) = \begin{cases} 0 & x < 0 \\ 1/6 & 0 \leq x < 2 \\ 1/2 & 2 \leq x < 4 \\ 5/8 & 4 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$.

Find $P(3 \leq x < 6)$, $M_X(t)$, $E(X)$ and $Var(X)$.

2. A random variable X has the CDF $F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\frac{x^2}{2\sigma^2}} & x \geq 0 \end{cases}$ where σ is a positive constant.

Find $P(\sigma \leq X \leq 2\sigma)$, $P(X > 3\sigma)$.

3. A random variable X has the following PDF, where $K > 0$: $f_X(x) = \begin{cases} 0 & x < 1 \\ K(x-1) & 1 \leq x \leq 2 \\ K(3-x) & 2 \leq x \leq 3 \\ 0 & x > 3 \end{cases}$. What is the value of K?

What is the CDF of X? What is $P(1 \leq X \leq 2)$.

4. Let X be a binomial random variable with mean 3 and variance 2, find $P(X = 2)$.

5. Find the mean and variance of the distribution whose moment generating function is $\frac{0.4e^t}{1-0.6e^t}$.

6. If X has uniform distribution in $(-1, 3)$ and Y has exponential distribution with parameter λ , find λ such that $Var(X) = Var(Y)$.

7. Let X be a continuous random variable with probability density function $f(x) = Kx^2e^{-x}$, $x \geq 0$. Find K, MGF of X and hence derive mean, variance.

8. Let X is a Poisson random variable. Find $P(X = 4)$ if $P(X = 1) = P(X = 2)$.

9. A man with 'n' keys wants to open his door and tries the keys independently and at random. Find the mean and variance of the number of trials required to open the door, (i) if unsuccessful keys are not eliminated from further selection, and (ii) if they are eliminated from further selection.

10. A certain machine makes electrical resistors having mean resistance of 40 ohms and variance of 4 ohms. Assuming that the resistance follows a normal distribution and can be measured to any degree of accuracy (i) What percentage of resistors have resistance that exceed 43 ohms? (ii) What percentage of resistors will have resistances between 35 ohms and 45 ohms?

11. A random variable X has the probability function $P_X(x) = \frac{1}{2^x}$, $x = 1, 2, 3, \dots$. Find its Variance.

12. Establish the forgot fullness property of (i) Geometric (ii) Exponential distributions.

13. Derive the MGF, mean and variance of (i) Binomial (ii) Poisson (iii) Geometric distributions.

14. Derive the mean and variance of (i) Uniform (ii) Exponential (iii) Gamma (iv) Weibull distributions.

15. Show that the sum of two normal random variables is a normal random variable.

16. If X is an exponential random variable with mean 2. Find the density function of $Y = \log X$.

17. Consider writing onto a computer disk and then sending it through a certifier that counts the number of missing pulses. Suppose this number X has a Poisson distribution with mean 0.2. What is the probability that a disk has exactly one missing pulse? What is the probability that a disk has at least two missing pulses? If two disks are independently selected, what is the probability that neither contains a missing pulse?

18. Let X denotes the number of grams of hydrocarbons emitted by an automobile per mile in a certain city. Assuming X is normal with mean 1 gram and standard deviation 0.25 gram. If ten automobiles are selected randomly from the city what is the probability that at least two of them will emit between 0.9 and 1.54 gram of hydrocarbons per mile?

19. Out of 500 families with 4 children each, how many families would be expected to have (i) at least 1 boy (ii) children of both sexes and (iii) at most 2 girls. Assume the probabilities for boys and girls are $1/3$ and $2/3$ respectively.

20. A fair die is thrown several times. Find the probability that a 4 will appear before 3 or 5.

21. The daily consumption of bread in a hostel, in excess of 2000 loaves is approximately Gamma distributed with parameters $k=1$ and $\lambda=1/1000$. The hostel has a daily stock of 3000 loaves. What is the probability that the stock is insufficient on a day?

24. Engineers designing the next generation of space shuttles plan to include two fuel pumps one active, the other in reserve. If the primary pump malfunctions, the second is automatically brought on line. Suppose a typical mission is expected to require that fuel be pumped for at most 50 hours. According to the manufacturer's specifications, pumps are expected to fail once every 100 hours. What are the chances that such a fuel pump system would not remain functioning for the full 50 hours?

25. Show that sum of two normal random variables is also a normal random variable.